

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION
January, 2020

Probability (Ph.D. Version)

Instructions to the Student

- a. Answer all six questions. Each will be graded from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your code number (**NOT YOUR NAME**) on the outside cover.
 - c. Keep scratch work on separate pages in the same booklet.
 - d. If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.
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Problem 1. A fair coin is tossed $n + 2$ times. For every $i = 1, \dots, n$ we define

$$\xi_i = \begin{cases} 1, & \text{if the outcome of the } i\text{-th and the } (i+2)\text{-th toss both differ from the} \\ & \text{outcome of the } (i+1)\text{-th toss,} \\ 0, & \text{otherwise.} \end{cases}$$

Let $\eta := \xi_1 + \dots + \xi_n$. Compute $\mathbb{E}(\eta)$ and $\text{Var}(\eta)$.

Problem 2. Let $X_1, X_2, \dots$ be a sequence of independent random variables such that $\mathbb{E}X_i = \mu$, for every i and $\sup_i \text{Var}(X_i) < \infty$. Let $Y_k = X_k X_{k+1}$ for every k . Prove that $\frac{1}{n} \sum_{k=1}^n Y_k$ converges to μ^2 in probability.

Problem 3. Let $(\xi_n)_{n \in \mathbb{N}}$ be a sequence of independent random variables defined as follows

$$\mathbb{P}(\xi_1 = 1) = \mathbb{P}(\xi_1 = -1) = \frac{1}{2},$$

and, if $n > 1$,

$$\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2\alpha},$$

$$\mathbb{P}(\xi_n = n) = \mathbb{P}(\xi_n = -n) = \frac{1}{2n^2} \left(1 - \frac{1}{\alpha}\right),$$

$$\mathbb{P}(\xi_n = 0) = 1 - \frac{1}{\alpha} - \frac{1}{n^2} \left(1 - \frac{1}{\alpha}\right),$$

for some constant $\alpha < 1$. Moreover, for every n and $1 \leq k \leq n$, define

$$\xi_{k,n} := \begin{cases} \xi_k, & \text{if } |\xi_k| \leq \sqrt{n}, \\ 0, & \text{if } |\xi_k| > \sqrt{n}. \end{cases}$$

1. If we define

$$S_n = \sum_{k=1}^n \xi_{k,n},$$

prove that

$$\sqrt{\frac{\alpha}{n}} S_n \xrightarrow{d} \mathcal{N}(0, 1).$$

2. Does the Central Limit Theorem hold for the sequence $(\xi_n)_{n \in \mathbb{N}}$?

Problem 4.

1. Let $(X_n)_{n \in \mathbb{N}}$ and $(Y_n)_{n \in \mathbb{N}}$ be two sequences of random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let Z be another random variable.

Prove that if

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \neq Y_n) = 0,$$

then

$$X_n \xrightarrow{d} Z, \text{ as } n \rightarrow \infty \iff Y_n \xrightarrow{d} Z, \text{ as } n \rightarrow \infty.$$

2. Let us define

$$T_n = \sum_{k=1}^n \xi_k,$$

where $(\xi_n)_{n \in \mathbb{N}}$ is the sequence introduced in Problem 3. Prove that

$$\sqrt{\frac{\alpha}{n}} T_n \xrightarrow{d} \mathcal{N}(0, 1).$$

Hint: You can use the first statement in this problem combined with the first statement in Problem 3, whether or not you have proved them.

Problem 5. Let $(\xi_n)_{n \in \mathbb{N}}$ be a sequence of independent random variables defined as follows

$$\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2^n}, \quad \mathbb{P}(\xi_n = 0) = 1 - \frac{1}{2^{n-1}}.$$

Let $\eta_n := \xi_1 + \dots + \xi_n$ and $\mathcal{F}_n := \sigma(\eta_k, k \leq n)$.

1. Prove that $(\eta_n)_{n \in \mathbb{N}}$ is a martingale w.r.t. the filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$.
2. Prove that $(\eta_n)_{n \in \mathbb{N}}$ is square integrable and determine the associated increasing process in the Doob decomposition of η_n^2 .
3. Does the martingale $(\eta_n)_{n \in \mathbb{N}}$ converge a.s.? In L^1 ?

Problem 6. Let W_t , $t \geq 0$, be a standard Brownian motion and let $s < t$. Compute the following

$$\mathbb{E}(W_s W_t^2), \mathbb{E}(W_s^2 W_t^2), \mathbb{E}(W_s e^{W_s}), \mathbb{E}(W_s e^{W_t}).$$